exoChile2026: Chilean Exoplanet Community Meeting

April 27-29, 2026, UCSC Concepción, Chile

Last updated: April 27, 2026

Day 1 Morning

8:30 — **Registration and arrivals.**

9:15 — **Welcome & Opening Remarks**.

Overview of objectives (*Matías Jones*) and format of the meeting (*Seba Pérez*)

9:30 — **Keynote** (*chair: Matías Jones*)

Andrés Jordán *Overview of Exoplanet Research from Chile*

10:10 — **Key Community Updates**. Short updates (7 min each) from different groups (*chair: Matías Jones*)

- David R. Anderson *La búsqueda de exoplanetas en tránsito liderada por la UCN*
- James Jenkins *Exoplanet research from UDP/CATA*
- Rafael Brahm *The WINE collaboration: Characterizing warm jupiters to solve the hot Jupiter puzzle*
- Jeremy Tregloan-Reed *Exoplanets in Chile International collaborations and northern state telescopes*
- Alice Zurlo *Exoplanet research from YEMS*
- *Gijs Mulders / Claudia Aguilera* *Exoplanets at PUC*
- Open time for updates

11:10 — **Coffee break**

11:40 — **"Vaciado": Community Mapping.** Each (sub-)community or groups identifies:

Current strengths — Tools, facilities, and datasets — Ongoing/planned projects — Gaps, needs, opportunities and dreams

12:45 — **Group presentations**

Day 1 Afternoon

14:30 — **Lightning Talks (3 min per talk)** *(chair: Máx Moyano)*

- H. Parashivamurthy TESS analysis of super-Earths around super small stars
- Ricardo Ramírez Time-domain Hα spectroscopy of wide-orbit planetary-mass companions with VLT/MUSE
- Michaela Vitkova Warm gas giants and their friends
- Scarlett E Royle The effect of stellar position-velocity clustering and galactic dynamics on exoplanet properties
- Ignacia Santiago Long-Term Dynamical Stability of Multiplanet Systems' Rocky Exoplanets in the Habitable Zone
- Andrea Bernardi Spectrophotometric analysis and orbital monitoring of directly imaged substellar companions
- Alejandra Fernández Autoencoders for Speckle Denoising in ADI Sequences
- Joanne Rojas Molina RV-Astrometry Synergy in the Detection of Warm Giants
- Ilaria Giovannini Looking for satellites with doppler measurements of directly imaged exoplanets and brown dwarfs
- Bárbara Rojas-Ayala Bayesian Model Averaging for Homogeneous Stellar and Exoplanet Characterisation from Late-K to Ultracool Dwarfs

16:00 — **Coffee break**

16:30 — **Contributed talks (10 min each)** *(chair: Nicola Astudillo-Defru)*

- Felipe Lagos-Vilches Observational Constraints on the Incidence of Substellar Companions to Young Massive White Dwarfs with VLT/ERIS
- Carolina Charalambous Lithium abundance in TOI-xxx: testing the planet-engulfment hypothesis
- Connor Cheverall Ground-based Atmospheric Characterization of Volcanic Super-Earth L98-59d at High Spectral Resolution
- Adam Rains VLT/CRIRES+ observations of warm Neptune WASP-107 b: Molecular detections and challenges in ground-based transmission spectroscopy of cooler and cloudy exoplanets
- Kevin Hoy First Evidence of Radial Velocity Signal from Satellite(s) Around a Substellar Companion
- Gijs Mulders Exoplanet Water Worlds: What are they and where can we find them?
- Claudia Aguilera-Gómez Polluted White Dwarfs as Probes of Exoplanetary Systems
- Jhon Yana Galarza What Can Alpha-Enhanced Stars Tell Us About Exoplanets?
- Marcela Best Kepler in Kepler: how an "alien" astronomer would develop Kepler's Third Law

Day 1 Evening

18:00 — **Community Icebreaker** (food/drinks)

19:00 — Public talk "Exoplanetas y vida en el espacio" by Alice Zurlo at Universidad Católica de la Santísima Concepción

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Day 2 Morning

9:30 — **Group presentations from Community Mappings. [NEW]**

10:00 — **Panel discussion [NEW]**

- AI for exoplanets — Connor Cheverall, Cecilia Hernández and James Jenkins

10:30 — **Mixed-Community Dialogues.** Participants rearrange into **diverse/mixed groups**, focusing on 1–2 questions, for example:

- Collaboration opportunities
- Chile's role in global exoplanet science
- Barriers to collaboration

11:00 — **Coffee Break**

11:30 — **Continuation of Mixed-Community Dialogues.**

12:00 — **Plenary Report-Back (~1 hr)**

- Groups summarize key ideas
- Emergent themes guide the afternoon's seed-lab topics

Day 2 Afternoon

14:00 — **Lightning talks (3 min each)** *(chair: Alice Zurlo)*

Fernando Castillo M.	Survival of vortices in gas-rich debris disks
Erick Montoya	Unveiling the real geometry of the FU Orionis system
Constanza Norambuena Herrera	Polluting white dwarfs: a dynamical study
Alex Mauch-Soriano	Predicted incidence of Jupiter-like planets around white dwarfs
Sidharth Aroli Veettil	The 14N/15N and 12C/13C isotopic ratios in the protoplanetary disk around V4046 Sgr
Constanza Castillo	The formation of formaldehyde (H2CO) in protoplanetary disks
Marie M. Rodríguez S.	Variability and Geometry of the Brγ Emission in HD 100546 from VLTI/GRAVITY
Milenne Avila Bravo	Planetary architectures under the influence of a stellar binary
Lucas Gonzalez-Rivas	Not Just Gas: How Solid-Driven Torques Shaped the Migration of the Galilean Moons
Grace Batalla Falcon	Evolution of the silicate feature in protoplanetary disks
Camilo González-Ruilova	Can the environment affect the planet formation?
Hannah Osborne	A homogenous analysis of small-planet masses

14:40 — **Contributed talks (10 min each)** *(chair: Gijs Mulders)*

Nicola Astudillo-Defru	Searching for terrestrial exoplanets with optimized radial velocities
Claudio Caceres	A search for free-floating planets in nearby young stellar associations
Neda Hejazi	Exploring Planetary Formation and Composition Using Detailed Chemical Abundances of Host Stars
Philipp Weber	Planet formation in the light of stellar accretion bursts
Matthias Schreiber	The importance of tides for the survival of planets during late stellar evolution

15:40 — **Collaboration Seed-Labs.** Self-organized breakout groups form around concrete ideas. For example:

- Joint proposals
- Shared data/AI tools or open repositories
- Training programs or student exchanges
- Outreach and public engagement initiatives

16:20 — **Coffee Break**

16:45 — **Seed-Lab Pitches (5 min/each).** Groups present concrete next steps and timelines.

17:15 — **Community Strategy Session** (Roadmap Discussion). Possibly as a small panel

- National priorities
- Future facilities
- Student training pathways

Day 2 Evening

19:00 — Public talk "Del Desierto de Atacama al cosmos" by Bárbara Rojas-Ayala at Universidad Católica de la Santísima Concepción

Day 3 Morning

9:45 — **Student-led reflection session.** A session *organized by students, moderated by postdocs.*

- Training needs
- Mentoring networks
- What barriers do students face in Chile?
- Community structure (how can the community better support early-career researchers?)
- Guided reflections on the need of national summer schools on exoplanet research, statistics and machine learning, or other topics
- Chilean exoplanet Slack workspace

10:45 — **From ideas to Working Groups**

11:00 — **Coffee Break**

11:30 — **Working Groups and Commitments.** Here's where we consolidate initiatives from seed labs and other sessions.

- Organizers compile “exoChile 2026 Community Initiatives”

12:00 — **Community Roadmap Discussion**

12:45 — **Next Steps and Closing Remarks**

- Who hosts exoChile2027 (or 2028?)
- Creation of communication channels

13:00 — **Farewell and Group Photo**

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List of Posters

ATM-01	Benjamin Gonzalez Gonzalez Vidal	Modeling of Archean Earth-like aquaplanet atmosphere: the effect of obliquity on climate and observables of exoplanets
ATM-02	Catalina Valesca Ávalos Vega	Breaking Degeneracies in Exoplanet Atmospheres: Measuring Rayleigh Slopes in exoplanets to Infer Mean Molecular Weight
DET-01	Sebastián Elias Cabrera Astudillo	BODIES: Big Or Dinky Investigation and Evaluation System
DET-02	Marcela Cristina López Ulloa	When M Dwarfs Flicker: Are Young Exoplanets Hiding There?
DET-03	Daniela Gonzalez	The Synergy of Gaia DR4 and VLTI/GRAVITY+: Low-Mass exoplanet detection in Binary Systems
DET-04	Joanne Michelle Rojas Molina	RV-Astrometry Synergy in the Detection of Warm Giants
DET-05	Florian Rodler	Mass determination of the three long-period Neptune- and sub-Neptune-sized planets transiting TOI-282
DET-06	Cathalina Andrea Zamora Fernández	Discovery and characterization of a long period system
DET-07	Franshezka Dominik Paz Tapia Candia	The discovery and characterization of a long period system
DET-08	José Luis Ricra Mayorca	Orbital period refinement and analysis of central transit time variations of WASP-50 b using ground-based and space observations
DET-09	Jose Vines	TIC 275647095 b, an eccentric Warm Jupiter Orbiting a Solar-analog Star
DET-10	Maximiliano Moyano	Discovery and characterization of a long period system
DISK-01	Hernán René Salinas Castro	Searching for Complex Organic Molecules with JWST–MIRI in Planetesimal Formation Regions
DISK-02	Abigali Rodríguez Jiménez	Rotation curves of prototype protoplanetary discs
EDU-01	Sol Francisca Troncoso Barrías; Lindsay Rincón Torres; Thiare Antonella Molina Tapia	Introduction to Data Science through the Exploration of Exoplanets as an Educational Strategy
INST-01	Oscar Castillo Vega	An optimized implementation of the PACO algorithm.
INST-02	Ian Rickmers Blamey	Deepermoments: Deep Learning for moment maps
OUT-01	Martina Pascall Molina Riffo; Bárbara Scarlett Quilodrán Hijerra; Javiera Cajas Reyes	Astronomical Collage as a Science Communication Tool with a Gender Perspective in Exoplanet Science Outreach
SYS-01	Felipe Ignacio Rojas Henríquez	Giant companions in Super-Earth systems from TESS
SYS-02	Álvaro Ignacio Llancaqueo Albornoz	Unveiling the nature of TESS giant planets in the first PLATO Field
SYS-03	Bhavya Ambalapatta kannancheri	Is the Hot Neptune Desert Really Empty? A Comparison of TESS and Kepler.

Poster Code Legend (Numbers indicate individual posters within each thematic category.)

DET — Detection & Characterization of Exoplanets
ATM — Exoplanet Atmospheres & Climate
SYS — Planetary Systems & Demographics
DISK — Protoplanetary Disks & Planet Formation
INST — Instrumentation, Methods & Algorithms
DS — Data Science, Machine Learning & Education
OUT — Outreach, Communication & Society

Your Role as Facilitator

You are not just moderating; you are **enabling/fostering collaboration**. Your responsibilities:

- Keep discussions **focused and on time**
- Ensure **everyone participates briefly**
- Identify **connections between participants**
- [Desirable but not compulsory] Push groups toward some **concrete outcomes**

General Principles (apply throughout all sessions)

- Prioritize **clarity over completeness**
- Avoid long monologues, better to have **short interventions**
- Actively ask: **"What do you need?" and/or** What do you need to be happy/productive? Reach your goals? What are your goals?
- Look for:
    - Shared datasets or tools
    - Complementary expertise
    - Overlapping scientific questions
- Push toward **smaller groups** when natural connections appear
- [Desirable but not compulsory] Steer toward **something actionable**

Day 1 — Community Mapping ("Vaciado", 11:40, duration 1hr)

We will work in different **"mesas de trabajo" / Work groups**. LOC will communicate locations to facilitators via Slack.
Groups are identified by the **color of the sticker** on your name badge.

Step 1 — [very important] Rapid Introductions (suggested duration ~20 min)

Each participant briefly states:

- Name & institution
- Research focus
- What they are currently working on
- **What they need / are looking for** (this is key).

Keep this **short (~1-2 minute per person)**

Step 2 — Live Mapping (~15 min)

As people speak, the facilitator should try to **capture patterns**, such as:

- Common research themes
- Shared datasets or instruments
- Similar methods
- Complementary needs

Step 3 — Trigger Connections (~15 min)

Use guiding questions:

- Who needs data/expertise that someone else has?
- Who is working on similar problems?
- Are there theory–observation complementarities?
- Could this become a paper or proposal?

Step 4 — [desirable but not compulsory] Form Small Clusters

Encourage spontaneous **subgroups (2–4 people)** to emerge around:

- Shared interests
- Complementary skills
- Common goals

Output of this session: A **clear map of the community. And hopefully** initial **clusters of potential collaboration**

Day 2 — Mixed-Community Dialogues (10:30, duration 1 hr)

Now participants are **re-mixed across communities**.

Your goal here: expand and challenge ideas, activate student participation if it doesn't happen naturally (ex: by asking what they know about other subfields or what interests them most).

Guide discussion around:

- Collaboration opportunities across subfields
- Chile's role in global exoplanet science
- Barriers (technical, institutional, cultural)

Push beyond discussion, and don't forget to GRAB A COFFEE (coffee break actually *breaks* the session in two)

Encourage participants to:

- Connect ideas from Day 1 clusters
- Identify **new interdisciplinary opportunities**
- Refine or rethink collaboration ideas

Key facilitator action: If promising ideas emerge: Encourage those people to **stay connected for the afternoon Seed-Labs**

Day 2 — Seed-Labs (15:40, duration ~40 min)

This is where ideas *could* become **concrete collaborations**. Facilitators can circulate between Seed-Lab groups, acting as catalysts to help those ideas become collaborations.

Step 1 — Form Groups Around Ideas

Participants self-organize into groups based on:

- Ideas from Day 1 or morning discussions
- New collaborations emerging during the meeting

Step 2 — Guide Toward a Concrete Plan

Each group should define (there will be a printed template/form to complete with this info):

- **What are we doing?** (clear problem/topic)
- **Who is involved?**
- **Why this collaboration?**
- **What is the concrete outcome?** Paper? Proposal? Shared dataset/tool? Training initiative?

Step 3 — Define First Steps

Push groups to specify:

- First actions
- Rough timeline
- Communication method

If there is no next step, it is not a real collaboration.

Day 2 — Seed-Lab Pitches (short summaries) (16:45)

Help groups prepare a **clear, concise pitch (5 min max)**:

They should answer:

- What is the idea?
- Who is involved?
- What will you produce?
- What are the next steps?

Encourage clarity and realism over ambition.